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Assignment 1

Analysis of a Cybersecurity Incident

Assigned Topic

Ransomware

Case Study: SafePay Ransomware Attack on Ingram Micro (July 2025)

Namisa Najah Raisa

210042112

Program: B.Sc in SWE

Department of CSE

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Abstract

Ransomware is malicious software that encrypts a victim's systems or steals data, demanding payment to restore access or prevent public exposure. This report analyses the July 2025 attack on Ingram Micro by the SafePay ransomware group, which caused a six-day global operational blackout and an estimated approximately USD 136 million in daily revenue losses.

Analysis of a Cybersecurity Incident

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Assigned Category: 3. Ransomware

(calculation: last two digits of student ID = 12 $\rightarrow$ 12 % 10 = 2 $\rightarrow$ +1 = 3)

1 Brief Summary

In July 2025, Ingram Micro, a leading business to business technology distributor with over 161,000 customers, 23,500 employees, and USD 48 billion in net sales in 2024, experienced a major ransomware attack attributed to the SafePay group.

The attack began on July 2 and 3, 2025. It encrypted internal systems and triggered a global operational outage that lasted several days. Core services, including the company's website, ordering platforms, and distribution operations, were brought to a halt. The timing, just before the U.S. Independence Day holiday weekend, appears deliberate, taking advantage of reduced staffing to increase impact.

SafePay gained access through the GlobalProtect VPN using stolen or brute forced credentials. The absence of multi factor authentication made this entry point viable. Once inside, the attackers moved laterally across the network, disabled key security tools such as Sophos and Veeam backups, and exfiltrated approximately 3.5 terabytes of sensitive data. Ransomware was then deployed across systems, with notes left on employee workstations as a final signal of compromise.

The incident resulted in estimated daily revenue losses of USD 136 million. It also exposed the personal data of 42,521 individuals, including employees and job applicants. Ingram Micro did not disclose whether a ransom was paid. However, the release of stolen data on SafePay's dark web site in early August 2025 suggests that negotiations, if any, did not end in payment. Global operations were fully restored between July 9 and 10, 2025.

2 Incident Title and Date or Period

Incident Title	SafePay Ransomware Attack on Ingram Micro: Data Breach and Extortion
Initial Intrusion	July 2, 2025 (data exfiltration began July 2 to 3)
Ransomware Deployment	July 3, 2025 (ransom notes appeared on employee systems)
Systems Disrupted	July 3 to 5, 2025 (Xvantage, Impulse, and ordering platforms)
Public Disclosure	July 5, 2025 (SEC Form 8-K press release)
SafePay Dark Web Claim	Late July 2025 (3.5 TB data leak announced)
Data Leaked publicly	Early August 2025 (stolen data published)
Operations Restored	July 9 to 10, 2025 (full recovery confirmed)
Breach Notification	January 2026 (Maine AG filing : 42,521 individuals notified)
Target Organization	Ingram Micro Holdings Inc., Irvine, California, USA
Sector	Technology distribution/IT supply chain
Threat Actor	SafePay ransomware group (active since September 2024)
Attack Type	Ransomware with double extortion
Data stolen	Approximately 3.5 terabytes (Claimed by SafePay)
Individuals Affected	42,521 individuals including employees and applicants
Estimated Daily Loss	Approximately USD 136 million

Table 1: Incident timeline

3 Detailed Description of the Event

Chronological Account

Prior to July 2, 2025 → Initial Access via Compromised VPN Credentials:

SafePay gained access to Ingram Micro's corporate network through the GlobalProtect VPN platform. Attackers used stolen or brute forced credentials to authenticate, taking advantage of the lack of enforced multi factor authentication. This aligns with SafePay's typical approach of targeting VPN gateways and exposed Remote Desktop Protocol endpoints where authentication controls are weak. The exact method used to obtain the credentials remains unclear, though phishing, password spraying, or initial access brokers are all plausible.

July 2 to 3, 2025 → Lateral Movement, Defence Evasion, and Exfiltration:

After gaining access, the attackers conducted internal reconnaissance using **ShareFinder.ps1**, a PowerShell script that identifies accessible network shares and high value data locations. They moved across the network using PsExec, Windows Remote Management, and RDP. These are legitimate administrative tools, which allowed the activity to blend in with normal operations.

The group then focused on weakening defences. Windows Defender was disabled using PowerShell commands and built in utilities. Security tools such as Sophos, backup services like Veeam, and critical system processes were terminated. Volume Shadow Copies were deleted to prevent recovery, and event logs were cleared to limit forensic visibility.

During this period, SafePay exfiltrated about 3.5 terabytes of data. Files were archived using WinRAR and transferred through FileZilla. Encryption was triggered only after the data had been secured, reflecting a deliberate double extortion strategy.

July 3, 2025 → Ransomware Deployment and Discovery:

On the morning of July 3, employees began seeing ransom notes named `readme_safepay.txt` on their systems. Files were encrypted and renamed with the **.safepay** extension. By around 8:00 AM ET, core systems started failing. Customers worldwide experienced outages as the company's website and ordering platforms became unavailable.

July 3 to 5, 2025 → Containment and Initial Response:

Ingram Micro responded by taking affected systems offline, including the Xvantage platform and the Impulse cloud licensing service. Employees were asked to work remotely where possible. The company engaged an external cybersecurity forensics firm and notified law enforcement.

On July 5, Ingram Micro publicly disclosed the incident through an SEC Form 8-K filing. The statement confirmed the presence of ransomware on internal systems and outlined initial containment and investigation efforts.

July 6, 2025 → Attacker Identification:

Media reports, including those from BleepingComputer, attributed the attack to the SafePay group. The reports reinforced that access was gained through compromised VPN credentials rather than a software vulnerability. Major outlets such as Reuters began covering the incident the same day.

July 7 to 9, 2025 → Staged Recovery:

Ingram Micro's teams worked around the clock to restore systems. Recovery was methodical and layered: beginning July 8, the company re-enabled subscription-based orders, then progressively restored phone and email ordering in specific countries before bringing full online systems back online. By July 9, Ingram Micro announced that global operations had been restored across all countries and regions.

Late July to August 2025 → Data Leak and Breach Notification:

SafePay listed Ingram Micro on its dark web leak site in late July, threatening to release the stolen data. The data was published in early August, which suggests that a ransom was not paid.

In January 2026, Ingram Micro filed a breach notification with the Maine Attorney General's Office. The filing confirmed that 42,521 individuals were affected, and notification efforts were initiated.

4 Root Cause / Vulnerability Exploited

The breach resulted from multiple security gaps rather than a single technical flaw. SafePay did not rely on a zero day vulnerability. Instead, the attack was built on compromised credentials and the misuse of legitimate administrative tools.

- **Compromised VPN Credentials as the Entry Point:** SafePay accessed Ingram Micro's network through the GlobalProtect VPN using stolen or brute forced credentials. The most likely methods include password spraying against the VPN gateway or credentials obtained from an initial access broker. Palo Alto Networks, the vendor of GlobalProtect, noted that such attacks commonly exploit stolen credentials or misconfigured access controls rather than software flaws.
- **Lack of Multi Factor Authentication on VPN Access:** The VPN did not enforce multi factor authentication. This meant a single compromised password was enough to gain full access. SafePay is known to target VPN and RDP gateways where authentication controls are weak. In this case, the absence of MFA turned a routine credential compromise into a full scale breach.
- **Weak Protection Against Security Tool Disabling:** SafePay was able to terminate Sophos endpoint protection and Veeam backup services before deploying ransomware. The attackers disabled Windows Defender using PowerShell and built in system tools. They also deleted Volume Shadow Copies and cleared event logs. These actions were completed without interruption, indicating a lack of tamper protection or hardened endpoint controls.
- **Limited Detection of Lateral Movement and Data Exfiltration:** After entry, the attackers moved across the network for at least a day using tools such as PsExec, WinRM, RDP, and ShareFinder.ps1. These tools are legitimate, which made detection more difficult. However, no system flagged large scale network enumeration, privilege escalation, or the transfer of several terabytes of data. The absence of behavioural monitoring and data loss prevention controls allowed the activity to proceed unnoticed until ransomware was deployed.

- **Insufficient Network Segmentation:** The attackers were able to move across systems with little resistance. This suggests that network segmentation was limited or ineffective. In a more segmented environment, access from a VPN connected endpoint would not easily extend to critical servers. The ability to reach and disable Veeam backup infrastructure shows that backup systems were not properly isolated from the main network.

5 Damage Caused

5.1 Operational Damage

The ransomware attack caused a near-total operational blackout across Ingram Micro's global business for approximately six to seven days. The company's website and all online ordering portals went offline. Its AI-powered Xvantage distribution platform and Impulse cloud licensing portal, the primary digital interfaces for its 161,000+ customers, were both taken offline. Partners and resellers from major vendors including Dell, Cisco, HPE, and Microsoft faced order processing delays and lost visibility into shipments and inventory precisely as quarterly sales periods were closing. Managed service providers dependent on Ingram Micro for cloud licence provisioning were similarly unable to fulfil customer commitments. The disruption created a ripple effect across the global IT supply chain, illustrating how the compromise of a central distributor amplifies into thousands of downstream business failures.

5.2 Financial Damage

Analysts estimated revenue losses exceeding USD 136 million per day during the operational blackout. Additional financial costs included engaging a third-party cybersecurity forensics firm for investigation and remediation; implementing enhanced monitoring and security controls across the environment; notifying 42,521 affected individuals and providing 24 months of complimentary credit monitoring and identity theft protection services; filing regulatory breach notifications across multiple jurisdictions; and potential civil liability from affected customers, partners, and impacted individuals. Ingram Micro did not confirm whether a ransom was paid. The public leak of stolen data in August 2025 suggests no payment was made.

5.3 Data Privacy Damage

Ingram Micro confirmed in its January 2026 filing with the Maine Attorney General's Office that 42,521 individuals were affected. The compromised information included names, contact information, dates of birth, Social Security Numbers, passport numbers, driver's licence numbers, other government-issued identification numbers, and employment-related data including work evaluations and hiring records. Current employees, former employees, and job applicants were all affected. The public release of this data on SafePay's dark web leak site in August 2025 means the information remains accessible to criminal actors.

5.4 Reputational Damage

The incident attracted widespread coverage from major cybersecurity and technology publications including BleepingComputer, SecurityWeek, TechRadar, CSO Online, and Reuters. Ingram Micro's stock reportedly dropped approximately **4%** in the immediate aftermath as investors reacted to the confirmed disruption. The company's communication during the crisis

drew criticism: there was a near-total blackout in public communications during the first 36 hours, and its failure to proactively update partners and customers on restoration timelines eroded confidence. As one of the world's most critical IT distribution nodes, the prolonged outage raised serious questions about the resilience of supply chain infrastructure to ransomware threats.

5.5 Supply Chain Impact

Because Ingram Micro functions as an intermediary between manufacturers and thousands of resellers and MSPs, the attack's damage extended far beyond the company itself. Enterprise buyers faced order backlogs and uncertainty over shipments. Resellers could not process inventory or fulfil customer commitments. Cloud service providers relying on Impulse for licence provisioning were left unable to activate or renew services for their clients. This supply chain amplification effect is precisely what makes large IT distributors high-value ransomware targets.

6 Detection Method

Ingram Micro did not detect the SafePay intrusion through proactive security monitoring. The attack was discovered reactively, only after the ransomware had already encrypted systems and deployed ransom notes across the environment. This represents a complete failure of early detection across all phases of the attack.

On the morning of July 3, 2025, employees began reporting that ransom notes like pop-up messages reading **readme_safepay.txt** had appeared on their computer screens. Encrypted files had been renamed with the **.safepay** extension across affected systems. By 8:00 AM ET, the company's primary systems began failing in sequence. The detection therefore occurred at the final and most destructive stage of the attack: post-encryption, when the damage had already been done.

Critically, all preceding attack phases like initial VPN credential access, internal network reconnaissance using ShareFinder.ps1, multi-day lateral movement via PsExec and WinRM, privilege escalation, the deliberate disabling of Sophos and Veeam, deletion of Volume Shadow Copies, clearing of Windows event logs, and the exfiltration of 3.5 terabytes of data via FileZilla, all went entirely undetected. SafePay's deliberate use of legitimate Windows administrative tools (Living-off-the-Land techniques) made its activity blend with normal IT operations, bypassing signature-based detection. The clearing of event logs before encryption further eliminated forensic artefacts that might have enabled earlier detection.

Following the initial detection via ransom notes, Ingram Micro's security monitoring later flagged anomalous network activity around 20:00 UTC on July 3. But by this point, key portals were already being taken offline as triage had begun. It took the company approximately 36 hours from the appearance of ransom notes to officially confirm the ransomware incident in its SEC Form 8-K press release on July 5, a delay that drew criticism and contributed to the drop in its stock price.

7 Recovery Process

- **July 3, 2025 → Immediate Containment:** Upon the appearance of ransom notes and the onset of system failures, Ingram Micro immediately isolated affected systems to halt further lateral spread and encryption. The company prohibited use of the **GlobalProtect VPN** connection to prevent continued attacker access. Employees were directed to work from home where operations permitted. Critical platforms including **Xvantage** and **Impulse** were proactively taken offline to contain the spread.
- **July 3 to 5, 2025 → Forensic Investigation and Law Enforcement Notification:** Ingram Micro engaged a specialist third-party cybersecurity firm to lead the forensic investigation. Law enforcement was notified. The company worked to determine the initial access vector, map all systems reached by the attackers, and establish the full scope of data accessed and exfiltrated. On July 5, Ingram Micro issued a public statement via an SEC Form 8-K confirming the incident and advising that it had *“taken steps to secure its environment, launched an investigation with the assistance of cybersecurity experts, and notified law enforcement.”*
- **July 5 to 7, 2025 → Partial System Restoration:** Ingram Micro’s Xvantage website was restored, though ordering and licensing systems remained offline during the ongoing investigation. The company publicly apologized to customers and partners for the disruption and began providing limited updates on restoration progress. Some customers were directed to place orders via phone or email as online channels remained unavailable.
- **July 7 to 9, 2025 → Staged Global Recovery:** Beginning July 8, Ingram Micro initiated a layered recovery approach. First re-enabling subscription-based orders, then expanding phone and email ordering to additional countries, before bringing full online systems back online in phases. This staged approach ensured that systems were cleaned, tested, and verified before being reactivated to avoid re-infection from any residual malware. By July 9, the company announced that global operations had been fully restored across all countries and regions in which it transacts business.
- **Enhanced Security Measures Implemented Post-Attack:** As systems were brought back online, Ingram Micro stated it had *“implemented additional safeguards and monitoring measures to protect our network environment.”* While the company did not publicly detail the specific controls added, the incident pointed directly to the specific gaps that required remediation:
 - Enforcing MFA on GlobalProtect VPN
 - Hardening endpoint protection against tamper attacks
 - Implementing egress monitoring to detect large outbound data transfers
 - Segmenting backup infrastructure from production network credentials.
- **January 2026 → Breach Notification:** Ingram Micro filed a mandatory breach notification with the Maine Attorney General’s Office confirming that **42,521** individuals had been affected. Written notifications were sent to all impacted individuals. The company offered **24 months of complimentary credit monitoring and identity theft protection services** to all affected people.

8 Preventive Measures and Lessons Learned

8.1 Lessons Learned from This Incident

- **The absence of MFA on a VPN gateway is an open door for ransomware operators.** SafePay's entire intrusion was built on a single stolen or brute-forced VPN credential. Had MFA been enforced on GlobalProtect, possession of that credential alone would have been insufficient for access. Ingram Micro's own post-attack response included immediately prohibiting use of the GlobalProtect VPN, an implicit acknowledgement that the VPN configuration was the root cause. This is the single most consequential lesson of the incident.
- **Security tools that can be killed by ransomware provide false assurance.** SafePay systematically disabled Sophos endpoint protection and Veeam backup services before encrypting systems. This is a documented SafePay pre-encryption step, executed against a hardcoded list of security processes and services. Security tools must be configured with tamper protection enabled, requiring administrative re-authentication before processes can be terminated. Backup solutions must be placed on network-isolated infrastructure that cannot be reached using production Active Directory credentials.
- **Legitimate tools are invisible to signature-based detection.** Every phase of SafePay's post-access activity, ShareFinder.ps1, PsExec, WinRM, RDP, WinRAR, FileZilla; used tools that are present in or routinely used within enterprise environments. No malware was dropped. This is precisely why Living-off-the-Land attacks bypass traditional antivirus. Ingram Micro lacked the behavioural anomaly detection needed to flag unusual patterns: a single account enumerating thousands of SMB shares, bulk outbound file transfers of terabytes of data, and mass shadow copy deletion should each independently have triggered alerts.
- **Data exfiltration is the last warning before encryption.** SafePay consistently completes data exfiltration before triggering encryption, making anomalous outbound data transfer the final detection opportunity before irreversible damage occurs. The 3.5 terabytes moved via FileZilla in the hours before encryption began generated no alert. A properly configured data loss prevention (DLP) system monitoring large outbound transfers to unfamiliar external destinations would have detected this activity and triggered an immediate response.
- **A communication plan is as important as a technical response plan.** The Ingram Micro attack stood out for its total blackout in communication during the critical first 36 hours. Partners, resellers, and customers had no information on the scope of the outage or expected restoration timelines, causing compounding operational disruption across the supply chain. An incident response plan must include pre-defined communication templates, spokesperson assignments, and customer notification timelines, in addition to technical containment procedures.

8.2 Recommended Preventive Measures

- **Enforce MFA universally on all remote access pathways**, including VPN gateways, RDP (Remote Desktop Protocol) endpoints, and administrative portals. No remote access entry point should rely on a password alone. MFA is the single most effective control against credential-based ransomware intrusions and would have directly prevented this attack.
- **Deploy Zero Trust Network Access (ZTNA) to replace or supplement VPN**. Ensure that authenticated users receive access only to specific applications they are authorized to use, rather than broad network access. ZTNA eliminates the lateral movement opportunities that flat VPN access provides.
- **Enable tamper protection on all endpoint security tools** to prevent ransomware from terminating antivirus and EDR processes. Security products such as **Sophos, CrowdStrike, and Microsoft Defender** all offer tamper-protection settings that require privileged re-authentication before the agent can be stopped.
- **Isolate backup infrastructure from production network credentials**. Backup solutions such as **Veeam** must be network-segmented so that they are inaccessible using production Active Directory credentials. Implement the *3-2-1-1-0 backup strategy: three copies, two media types, one offsite, one immutable, and zero errors on the last verified restore test*.
- **Implement Data Loss Prevention (DLP) and anomalous egress monitoring** to alert on large outbound data transfers to unfamiliar external destinations. SafePay's exfiltration phase, terabytes transferred via FileZilla, represents the last window of detection before encryption. Outbound traffic monitoring should trigger automatic alerts on volume anomalies.
- **Deploy behavioural EDR with anomaly detection** configured to alert on specific SafePay-associated patterns: execution of ShareFinder.ps1, shadow copy deletion via vs-admin, mass PowerShell-based service termination, PsExec execution from non-standard accounts, and bulk file renaming events. Signature-based antivirus is insufficient against Living-off-the-Land techniques.
- **Develop and rehearse a comprehensive incident response plan** that explicitly includes a crisis communication playbook that includes partner and customer notification templates, spokesperson designations, and escalation timelines. The 36-hour communication blackout during the Ingram Micro attack significantly compounded the operational and reputational damage.
- **Conduct regular red team exercises targeting credential-based VPN intrusion paths** to verify whether a single compromised VPN credential can lead to lateral movement, privilege escalation, and data exfiltration. Ingram Micro's breach was entirely credential-based; this vector should be the primary focus of any penetration test for organisations with large VPN-connected environments.

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